

GSM-Based LPG Gas Leak Detection & Auto Shut-off System

Arduino Source Code (.ino)

```
#include <Servo.h>
#include <SoftwareSerial.h>

// ----- Pin Definitions -----
#define MQ2_PIN A0
#define SERVO_PIN 9
#define RELAY_PIN 10
#define BUZZER_PIN 6
#define LED_PIN 13

SoftwareSerial gsm(7, 8);    // RX, TX

Servo valve;

// Change this to your mobile number
String phoneNumber = "+91XXXXXXXXXX";

// Gas Threshold
int gasThreshold = 350;

bool alertSent = false;

//-----

void setup()
{
  Serial.begin(9600);
  gsm.begin(9600);

  pinMode(RELAY_PIN, OUTPUT);
  pinMode(BUZZER_PIN, OUTPUT);
  pinMode(LED_PIN, OUTPUT);

  digitalWrite(RELAY_PIN, LOW);
  digitalWrite(BUZZER_PIN, LOW);
  digitalWrite(LED_PIN, LOW);

  valve.attach(SERVO_PIN);
  valve.write(0);          // Valve OPEN

  delay(3000);

  initGSM();

  Serial.println("SYSTEM READY");
}

void loop()
{
  int gasValue = analogRead(MQ2_PIN);

  Serial.print("Gas Value : ");
  Serial.println(gasValue);

  if(gasValue > gasThreshold)
  {
    digitalWrite(RELAY_PIN, HIGH);
    digitalWrite(BUZZER_PIN, HIGH);
    digitalWrite(LED_PIN, HIGH);

    valve.write(90);        // Close Gas Regulator
```

```

        if(alertSent == false)
        {
            sendSMS("WARNING! LPG GAS LEAK DETECTED.");
            makeCall();

            alertSent = true;
        }
    }
else
{
    digitalWrite(RELAY_PIN, LOW);
    digitalWrite(BUZZER_PIN, LOW);
    digitalWrite(LED_PIN, LOW);

    valve.write(0);

    alertSent = false;
}

delay(1000);
}

//-----
// GSM INITIALIZATION
//-----

void initGSM()
{
    sendAT("AT",2000);
    sendAT("ATE0",2000);
    sendAT("AT+CMGF=1",2000);

    Serial.println("GSM Ready");
}

//-----
// SEND AT COMMAND
//-----

void sendAT(const char *command, unsigned long timeout)
{
    gsm.println(command);

    unsigned long time = millis();

    while(millis()-time < timeout)
    {
        while(gsm.available())
        {
            char c = gsm.read();
            Serial.write(c);
        }
    }

    Serial.println();
}

//-----
// SEND SMS
//-----

void sendSMS(String message)
{
    gsm.println("AT+CMGF=1");
    delay(1000);
    gsm.print("AT+CMGS=\"");

```

```
gsm.print(phoneNumber);
gsm.println("");

delay(1000);

gsm.print(message);

delay(500);

gsm.write(26);

delay(5000);

Serial.println("SMS SENT");
}

//-----
// MAKE CALL
//-----

void makeCall()
{
    gsm.print("ATD");
    gsm.print(phoneNumber);
    gsm.println(";");

    delay(15000);

    gsm.println("ATH");

    Serial.println("CALL COMPLETED");
}
```